Cache memories

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Computer Organization and Design, 4th Edition

What do programmers wish? (related to performance)

What do programmers wish?

Unlimited amounts of fast memory!

What do programmers wish?

1. Unlimited amounts of
2. Fast memory!

A concrete example

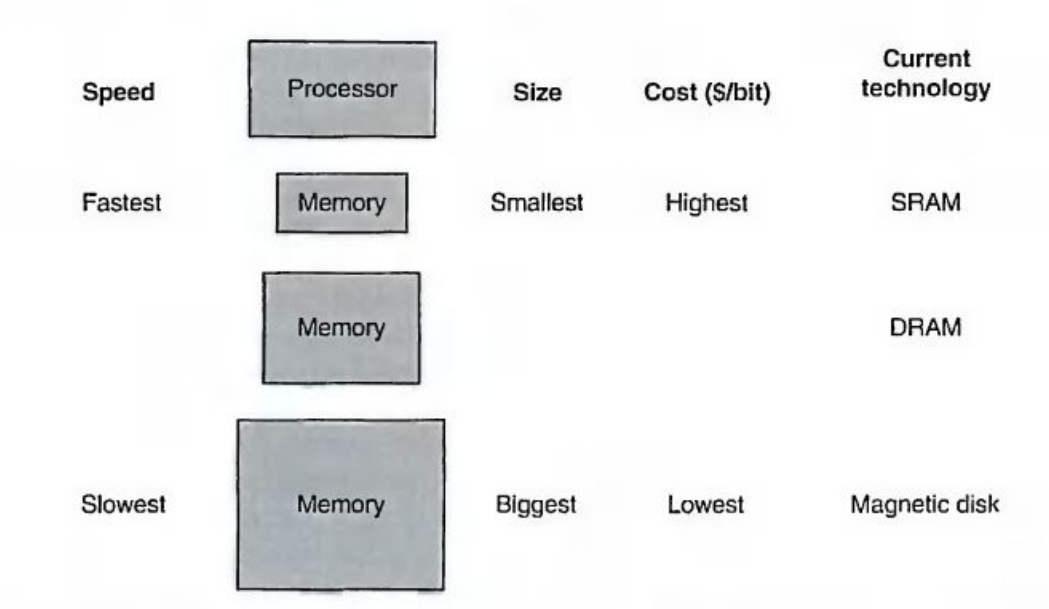
Locality principle

Temporal locality

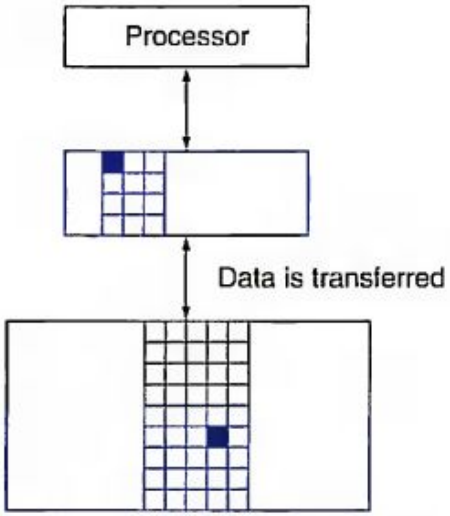
Locality principle

Spatial locality

Memory hierarchy



Memory hierarchy

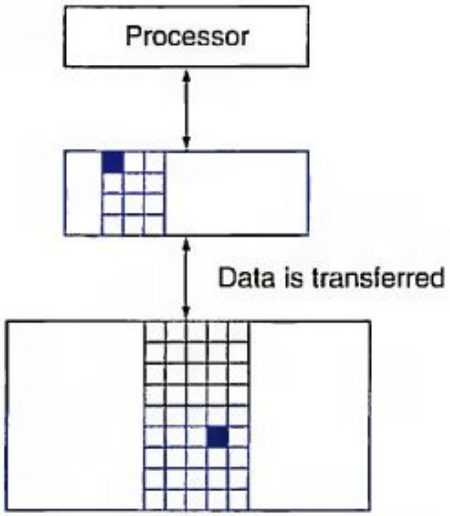


Quiz

Which of the following statements are generally true?

1. Caches take advantage of temporal locality.
2. On a read, the value returned depends on which blocks are in the cache.
3. Most of the cost of the memory hierarchy is at the highest level.
4. Most of the capacity of the memory hierarchy is at the lowest level.

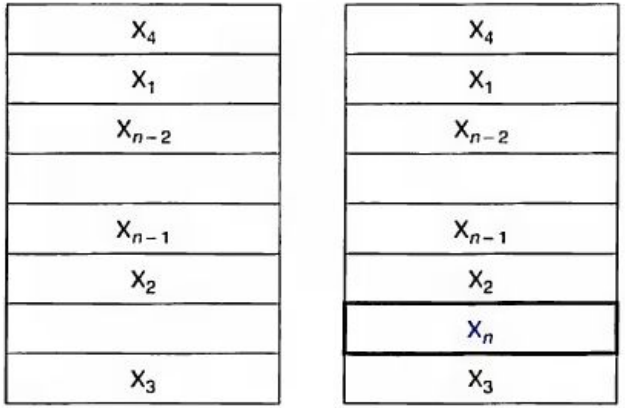
Memory hierarchy



Hit rate, hit time

Miss rate, miss penalty

Memory hierarchy



a. Before the reference to X_n b. After the reference to X_n

Hit rate, hit time

Miss rate, miss penalty

DM (Direct Mapped) Cache

Tag + block => index

Valid bit

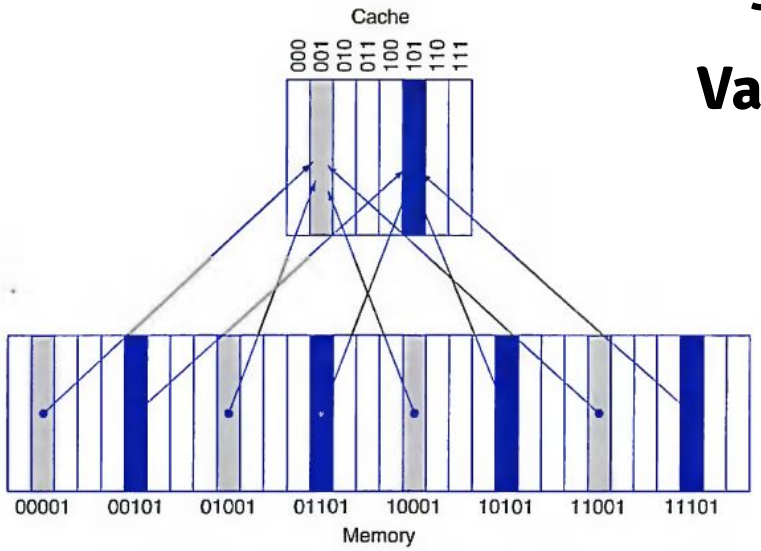


FIGURE 5.5 A direct-mapped cache with eight entries showing the addresses of memory words between 0 and 31 that map to the same cache locations. Because there are eight words in

Cache memory example

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	N		
111	N		

a. The initial state of the cache after power-on

Cache memory example

Index	V	Tag	Data
000	N		
001	N		
010	N		
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

b. After handling a miss of address (10110_{two})

Cache memory example

Index	V	Tag	Data
000	N		
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

c. After handling a miss of address (11010_{two})

Cache memory example

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	N		
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

d. After handling a miss of address (10000_{two})

Cache memory example

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	11_{two}	Memory (11010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

e. After handling a miss of address (00011_{two})

Cache memory example

Index	V	Tag	Data
000	Y	10_{two}	Memory (10000_{two})
001	N		
010	Y	10_{two}	Memory (10010_{two})
011	Y	00_{two}	Memory (00011_{two})
100	N		
101	N		
110	Y	10_{two}	Memory (10110_{two})
111	N		

f. After handling a miss of address (10010_{two})

Important formulas

n

Memory size = 2^n blocks

1 B = 8 b
(1 byte = 8 biți)

m

Cache size = 2^m blocks
m bits for index

Tag size

No. of bits/word = 32
32 - m - block offset - word offset

Important formulas

TAG	INDEX	BLOCK OFFSET	WORD OFFSET (only for byte addressed)
32-m-b.o.-w.o	m	$\log_2$(no. of **words / block**)	$\log_2$(no.of **bytes / word**)

n

Memory size = 2^n blocks

m

Cache size = 2^m blocks
m bits for index

Tag size

No. of bits/word = 32
32 - n - block offset - word offset

1 B = 8 b
(1 byte = 8 biți)

Word addressed vs. byte addressed

Problem no. 1

How many total bits are required for a D.M. cache with 64 KiB of data and 4-word blocks, assuming a 32-bit address? We consider a word has 32 bits and the memory is **byte** addressable.

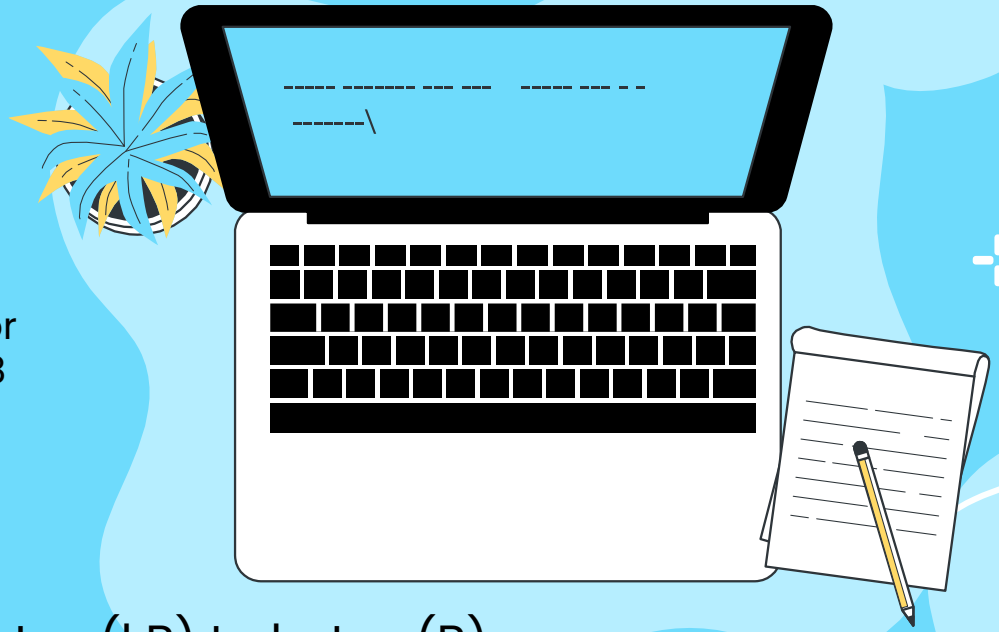
- Transform: data from kibibytes (kiB) to bytes (B)
- Compute: no. of words, no. of blocks, offsets (word offset, block offset), no. of bits for tag
- Multiply data for tag and **valid** and add it with data

Problem no. 2

How many total bytes are required for a D.M. cache with 32 KiB of data and 8-word blocks, assuming a 32-bit address? We consider a word has 32 bits and the memory is word addressable.

Homework

How many total bytes are required for a D.M. cache with 32 KB of data and 8 words/block, assuming a 32-bit address? We consider a word has 32 bits and the memory is byte addressable.



- Transform: data from kilobytes (kB) to bytes (B)
- Compute: no. of words, no. of blocks, offsets (word offset, block offset), no. of bits for tag
- Multiply data for tag and valid and add it with data

Thanks!